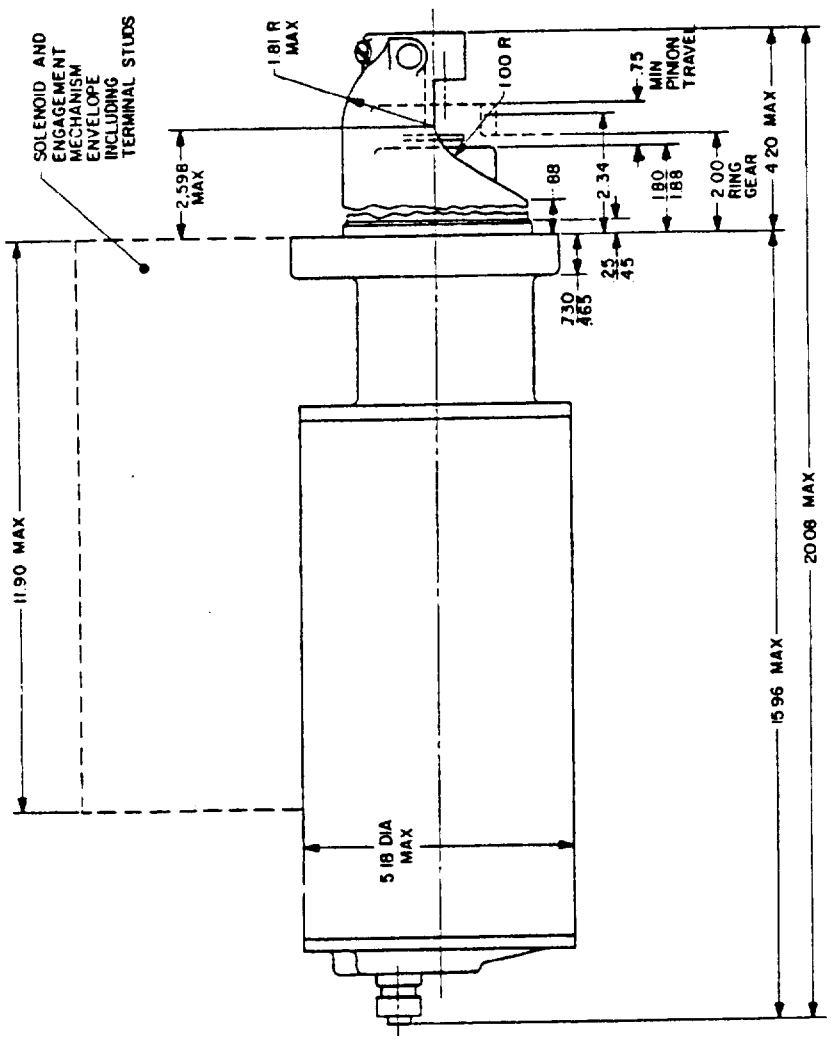
Review Activity: ARMY-AT

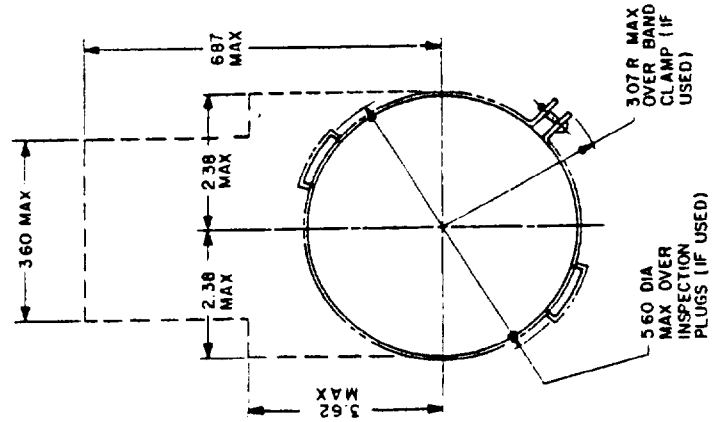
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MS PART NO.	TEETH	PITCH	MOUNTING POSITIONS
MS 53011-1	11T-12T BLANK	8/8	SEE SHEET 3
MS 53011-2	12T-13T BLANK	8/10	SEE SHEET 3
MS 53011-4	13T-14T BLANK	8/10	SEE SHEET 3



APPROVED 2 MAY 1962 REVISED (A) 28 JANUARY 1965 (B) 23 JUNE 1967 (C) 17 NOVEMBER 1970 (D) 2 JANUARY 1974 (E) 6 FEB 76

(E) SHEET 6 ADDED

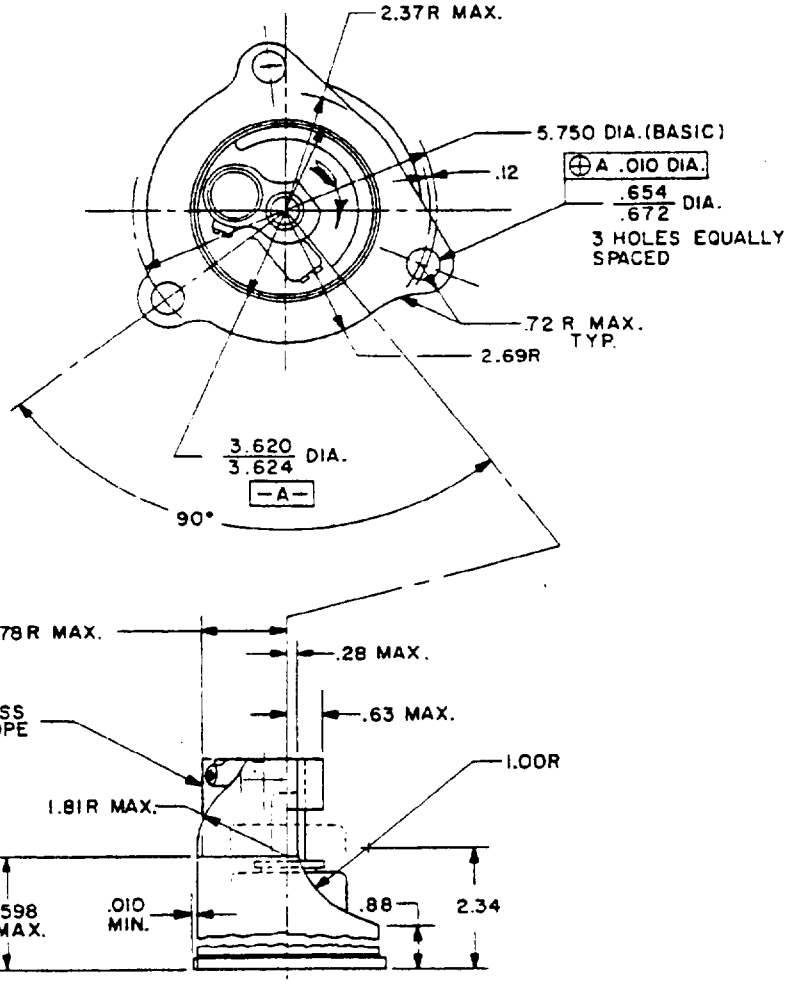
P.A. ARMY - ME Other Cust	TITLE STARTER, ENGINE ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY, LONG-FRAME	MILITARY STANDARD MS 53011
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES	SHEET 1 OF 6

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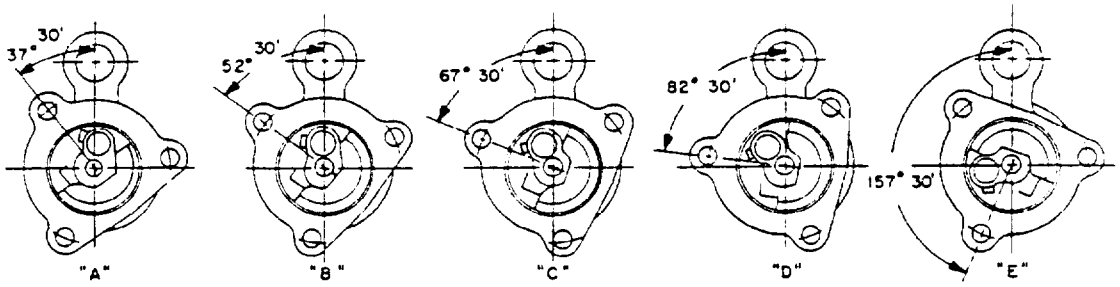
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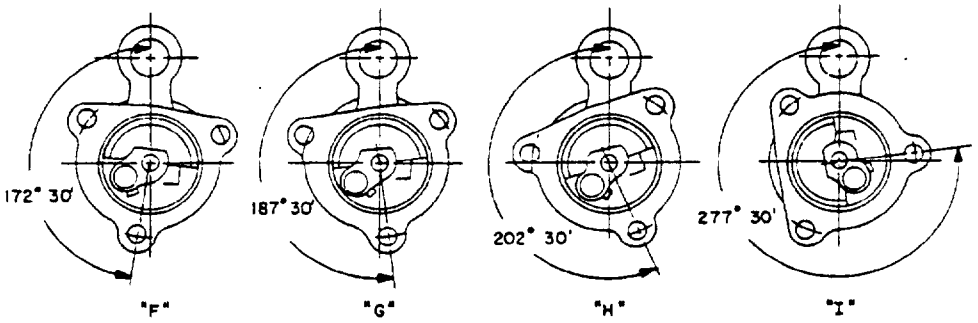
P.A. ARMY - ME Other Cust	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY, LONG FRAME	MILITARY STANDARD MS 53011
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES	SHEET 2 OF 6

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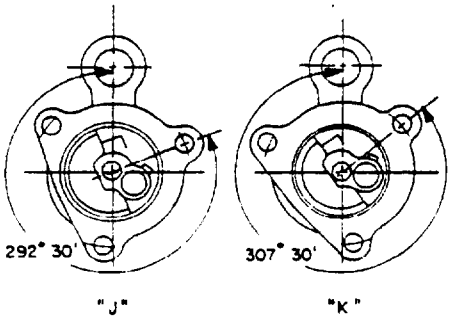
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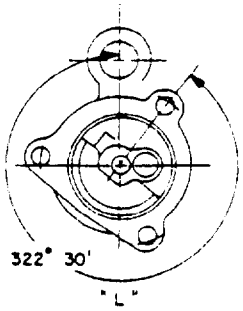
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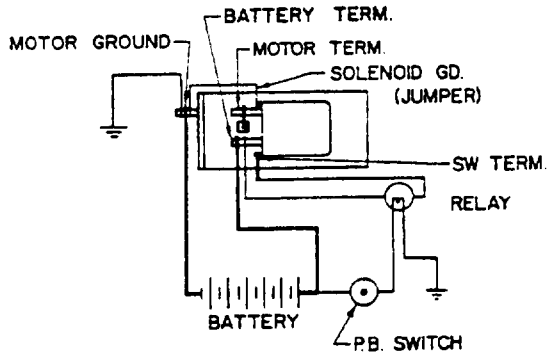




MOUNTING FLANGE AND PINION HOUSING FOR MS-1, MS-2, AND MS-4 STARTERS SHALL BE ADJUSTABLE TO ALL POSITIONS SHOWN HEREON. MOUNTING POSITIONS VIEWING DRIVE END OF MOTOR.







TYPICAL WIRING DIAGRAM (REVISED)

APPROVED 2 MAY 1962. REVISED (D) FOR CHANGES SEE SHEET 1 THRU 5 (E) FOR CHANGES SEE SHEET 6

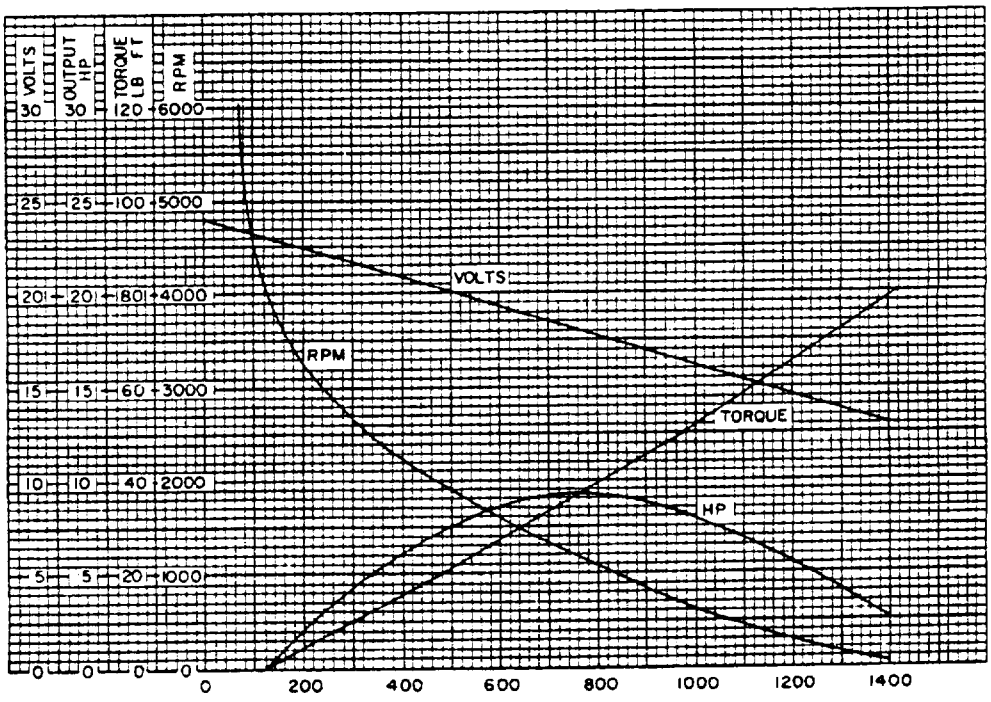
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P.A. ARMY - ME Other Cust	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT D.C., 5-1/8-INCH FRAME DIAMETER SOLENOID ACTUATED HEAVY-DUTY, LONG FRAME	MILITARY STANDARD MS 53011
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES	SHEET 3 OF 6

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PERFORMANCE CURVE

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P.A. ARMY - ME Other Cust	TITLE STARTER ENGINE, ELECTRICAL: 24 - VOLT D.C 5 - 1/8 - INCH FRAME DIAMETER SOLENOID-ACTUATED HEAVY-DUTY LONG -FRAME	MILITARY STANDARD MS 53011
PROCUREMENT SPECIFICATION MIL S-3785	SUPERSEDES	SHEET 4 OF 6

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NOTES:

1. PINION DATA: MS 53011-1 AND -3; 11 TEETH; 6/8 PITCH (12 TOOTH BLANK); 2.000 PD; 2.250 OD; 20° PRESSURE ANGLE; .015 TO .025 PL. CLEARANCE.
   MS 53011-2; 12 TEETH (13 TOOTH BLANK); 8/10 PITCH; 1.625 PD; 1.825 OD; 20° PRESSURE ANGLE; .015 TO .025 PL CLEARANCE.
   MS 53011-4; 13 TEETH; 8/10 PITCH (14 TOOTH BLANK); 1.750 PD; 1.950 OD; 20 DEGREE PRESSURE ANGLE; .015 TO .025 PL CLEARANCE.
2. MOTOR IS INSULATED TYPE.
3. MOUNTING IS SIMILAR TO #3 SAE FLANGE MOUNTING.
4. TERMINAL THREAD SIZE AND LENGTH: BATTERY -1/2-13 UNC 2A, MINIMUM LENGTH-.50, MAX 1.00 BEYOND OUTSIDE NUT WHEN JAM NUT, WASHER AND OUTSIDE NUT ARE ASSEMBLED ON TO THE STUD.
   MOTOR -GROUND- 1/2-13 UNC 2A, MAX. LENGTH .50 BEYOND OUTSIDE NUT WHEN STARTER IS ASSEMBLED.
   SWITCH -#10-32 UNF 2A (MS-1, MS-2 AND MS-4)
   -1/4-20 UNF 2A (MS-3)
5. DESIGN AND ARRANGEMENT OF SOLENOID AND ENGAGEMENT MECHANISM ASSEMBLY ARE OPTIONAL, PROVIDED SHIFT LEVER AND LINKAGE ARE ENCLOSED AND ALSO PROVIDED TERMINAL THREAD SIZES CONFORM TO AND LIMITING OVERALL DIMENSIONS DO NOT EXCEED THOSE SPECIFIED HEREIN.
6. ROTATION VIEWING DRIVE END - CLOCKWISE.
7. TOLERANCE: DIMENSIONAL: ± .030 UNLESS OTHERWISE SPECIFIED. ANGULAR: ±1°
8. THIS MOTOR IS IDENTIFIED AS TYPE II (RING GEAR ENGAGEMENT), GRADE A (WATERPROOF), CLASS 1 (POSITIVE DRIVE) OF REFERENCED SPECIFICATION.
9. MOTOR SHALL BE EQUIPPED WITH OIL SEAL FOR WET CLUTCH OPERATION.
10. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
11. MODIFICATIONS IN OUTSIDE APPEARANCE AND CONSTRUCTION ARE ACCEPTABLE IF UNIT INTERCHANGEABILITY IS NOT AFFECTED.
12. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
13. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
14. TORQUE AND R.P.M. MUST NOT BE LESS THAN 10% BELOW THAT SHOWN BY CURVES, WITH VOLTAGE APPLIED AS SPECIFIED. OUTPUT CURVE IS NOT RELATED TO TORQUE AND RPM CURVES SHOWN, BUT IS THE MINIMUM FOR ANY PARTICULAR MOTOR.
15. THE VOLTAGE CURVE SHOWN IS THE SAE HEAVY-DUTY 24-VOLT BATTERY CURVE.
16. LOCATION OF NAMEPLATE SHALL BE OPTIONAL. NAMEPLATE SHALL BE FASTENED TO STARTER FRAME WITH RIVETS OR SCREWS AND LOCK WASHERS.
17. STALLED TORQUE DATA: VOLTS - 3.5 (APPROX.)
    CURRENT (AMPS) (MAX) - 500
    TORQUE (LB FT) (MIN) - 16
18. JUMPER BETWEEN SOLENOID-GROUND TERMINAL AND MOTOR-GROUND TERMINAL SHALL BE FURNISHED WITH THIS STARTER.
19. SOLENOID CURRENT CONSUMPTION (PULL-IN PLUS HOLD-IN) SHALL NOT EXCEED 81 AMPERES.
20. WHEN THIS MOTOR IS SUPPLIED AS A REPAIR PART, THE MOVING FLANGE POSITION SHALL BE ORIENTED AS SPECIFIED IN PROCUREMENT DOCUMENTS. PACKAGED WITH EACH MOTOR WILL BE A DETAILED SHEET (AND/OR DRAWING), SHOWING THE PROPER MOUNTING FLANGE POSITIONS FOR OTHER APPLICATIONS AS NOTED IN THE PROCUREMENT DOCUMENTS. THIS DETAIL SHEET MUST ALSO INCLUDE INSTRUCTIONS FOR REPOSITIONING THE MOUNTING FLANGE AS REQUIRED.

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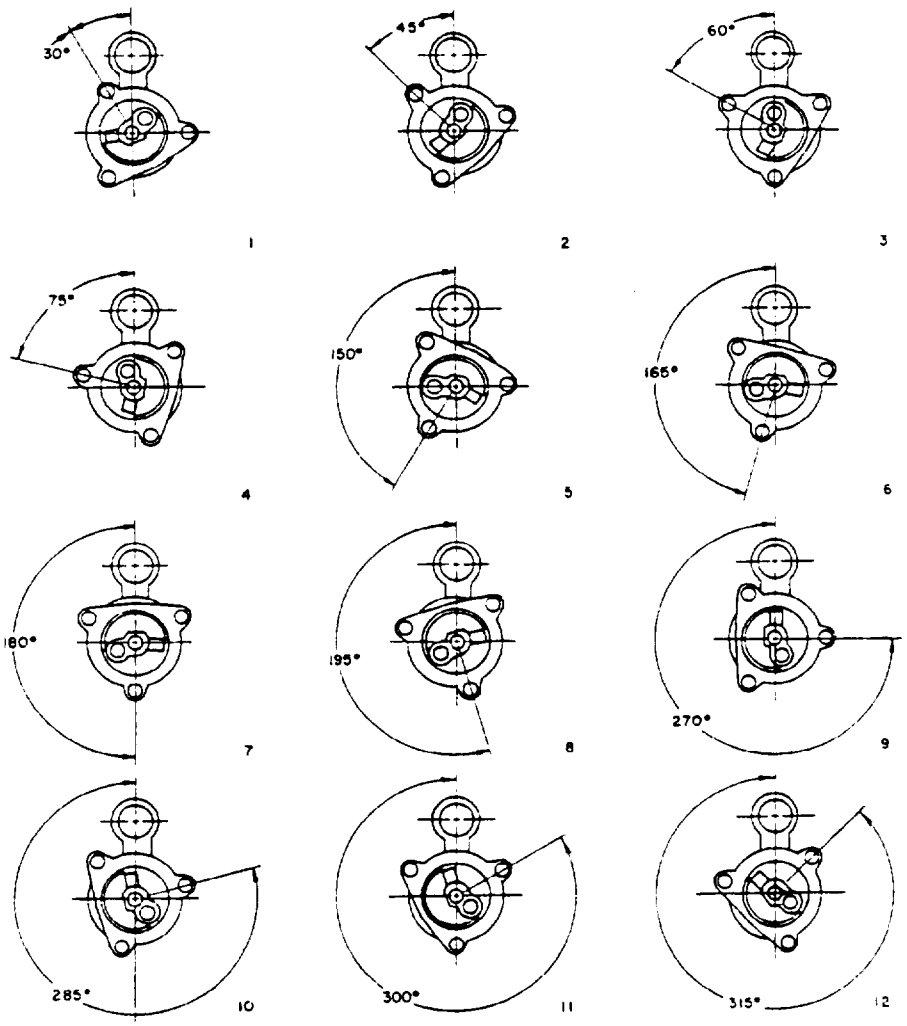
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P.A. ARMY - ME Other Cust	TITLE STARTER, ENGINE, ELECTRICAL: 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY, LONG-FRAME	MILITARY STANDARD MS 53011
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 5 OF 6

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POSSIBLE SWITCH LOCATION WITH RESPECT TO MOUNTING FLANGE, VIEWING DRIVE END.

REVIEW ACTIVITIES: ARMY - AT
USER ACTIVITIES: NAVY - MC

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P.A. ARMY - ME Other Cust	TITLE STARTER ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-IN. FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY, LONG FRAME.	MILITARY STANDARD MS 53011
PROCUREMENT SPECIFICATIONS: MIL-S-3785	SUPERSEDES:	SHEET 6 OF 6

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